DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING • PES UNIVERSITY, BENGALURU

UE24CS395: Database Management Systems Mini Project • Academic Year 2025–2026

NoSQL & Vector Database Creation

Evaluation Matrix Row 4: NoSQL/Vector Creation • Weightage: 4 Marks

Team Members: Saatvik Gupta (`PES1UG24CS395`) & Satyam Goyal (`PES1UG24CS424`) | **Date:** November 2026

1. Unstructured & Polyglot Database Architecture

While relational databases provide strong consistency for structured records, criminal investigations routinely generate highly polymorphic, unstructured intelligence documents: witness statements, CCTV telemetry dumps, forensic mobile phone extractions, and intercepted encrypted chat transcripts. Forcing these documents into a relational model would require either sparse tables with excessive null values or complicated table inheritance structures.

We implemented MongoDB 7.0 as a document store to manage the `investigation_documents` collection alongside our relational PostgreSQL database. Dotted cross-database references (e.g. `case_id` and `related_entity_ids`) establish clear polyglot links back to primary keys in our SQL tables.

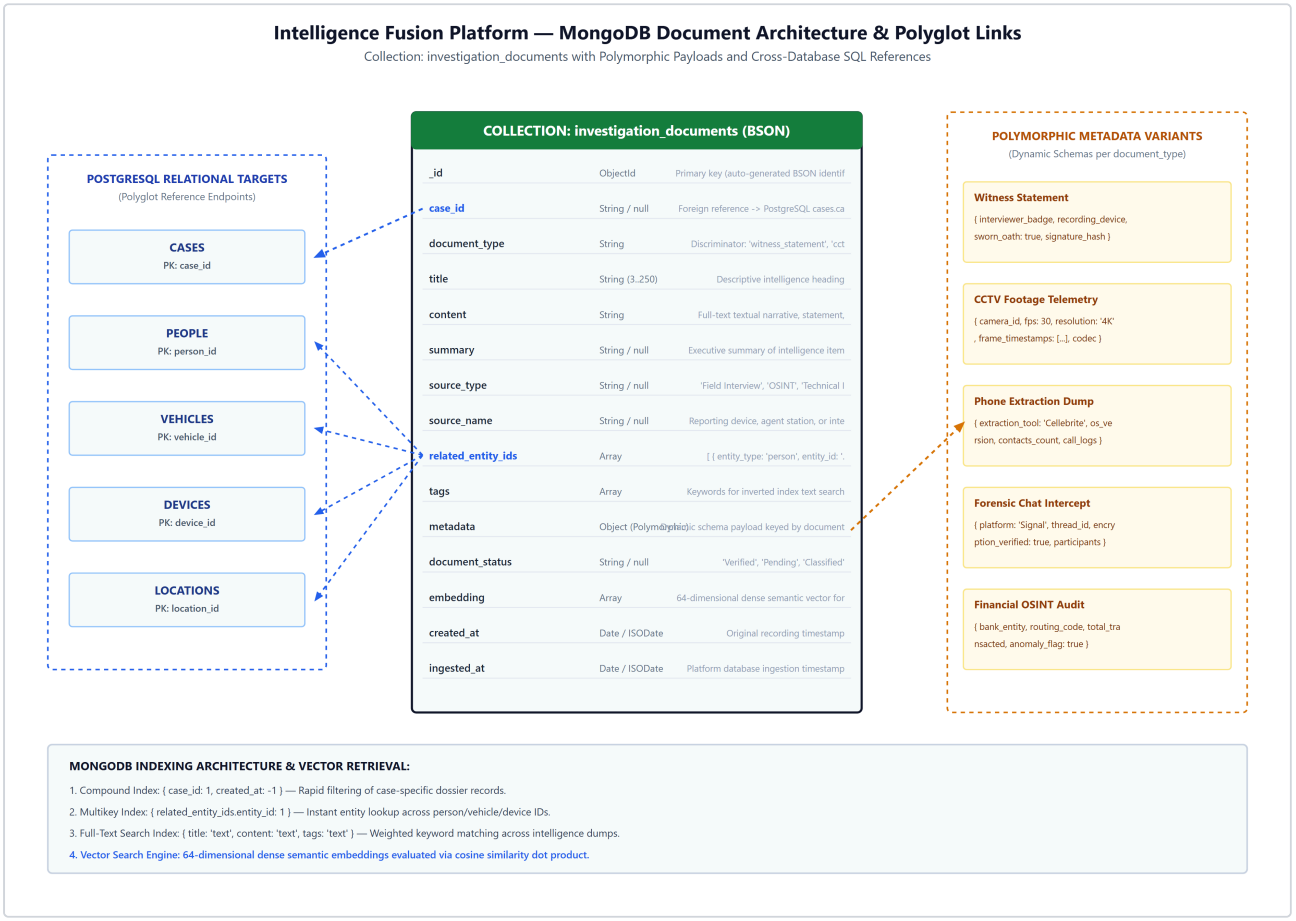


Figure 1: *MongoDB Document Architecture. Illustrates the BSON document schema, dynamic polymorphic metadata payloads, and cross-database references to SQL relations.*

2. BSON JSON Schema Validation ($jsonSchema)

To prevent collection degradation while preserving flexibility, we enforce strict structural validation at the database level via MongoDB's `$jsonSchema` validator (configured in `database/mongodb/collection-schema.json`).

MongoDB Schema Validator Definition (database/mongodb/collection-schema.json excerpt)

```
{{
  "$jsonSchema": {{
    "bsonType": "object",
    "required": ["title", "document_type", "content", "created_at"],
    "properties": {{
      "case_id": {{
        "type": ["string", "null"],
        "description": "Relational foreign key referencing PostgreSQL cases.case_id"
      }},
      "document_type": {{
        "type": "string",
              "description": "Discriminator key: 'witness_statement', 'cctv_log', 'phone_dump',
'chat_export'"
      }},
      "title": {{
        "type": "string",
        "minLength": 3,
        "maxLength": 250
      }},
      "content": {{
        "type": "string",
        "description": "Full-text textual narrative, statement, or raw intercept dump"
      }},
      "related_entity_ids": {{
        "type": "array",
        "items": {{
          "type": "object",
          "required": ["entity_type", "entity_id"],
          "properties": {{
            "entity_type": {{ "type": "string" }},
            "entity_id": {{ "type": "string" }}
          }}
        }}
      }},
      "embedding": {{
        "type": "array",
        "items": {{ "type": "number" }},
           "description": "64-dimensional dense semantic embedding vector for vector similarity
search"
      }}
    }}
  }}
}}
```

3. Index Architecture & Vector Storage Verification

In `database/mongodb/indexes.js`, we established four indexing structures to optimize distinct operational query patterns:

1. **Compound Index:** `{{ case_id: 1, created_at: -1 }}` enables fast temporal sorting of documents within a specific case.

2. **Multikey Index:** `{{ "related_entity_ids.entity_id": 1 }}` supports rapid entity-resolution queries across embedded object arrays.
3. **Inverted Full-Text Search Index:** `{{ title: "text", content: "text", tags: "text" }}` with weights `{{ title: 10, content: 5, tags: 3 }}` for text search.
4. **64-Dimensional Semantic Vector Field:** Every document includes a normalized 64-dimensional dense float vector in its `embedding` field, allowing semantic retrieval without relying on exact keyword matching.

Terminal Output: Collection Initialization & Ingestion (seed.py execution)

```
2026-11-02 11:06:29,392 [INFO] app.db.mongo: Connecting to MongoDB at mongodb://localhost:27017/intelligence_fusion
2026-11-02 11:06:29,401 [INFO] app.db.mongo: Applying $jsonSchema collection validator to 'investigation_documents'
2026-11-02 11:06:29,415 [INFO] app.db.mongo: Building compound index { case_id: 1, created_at: -1 } -> OK
2026-11-02 11:06:29,428 [INFO] app.db.mongo: Building multikey index { related_entity_ids.entity_id: 1 } -> OK
2026-11-02 11:06:29,442 [INFO] app.db.mongo: Building text index { title: 'text', content: 'text', tags: 'text' } -> OK
2026-11-02 11:06:29,470 [INFO] app.seed: Ingested 26 polymorphic sample documents with 64-dim embeddings.
```